

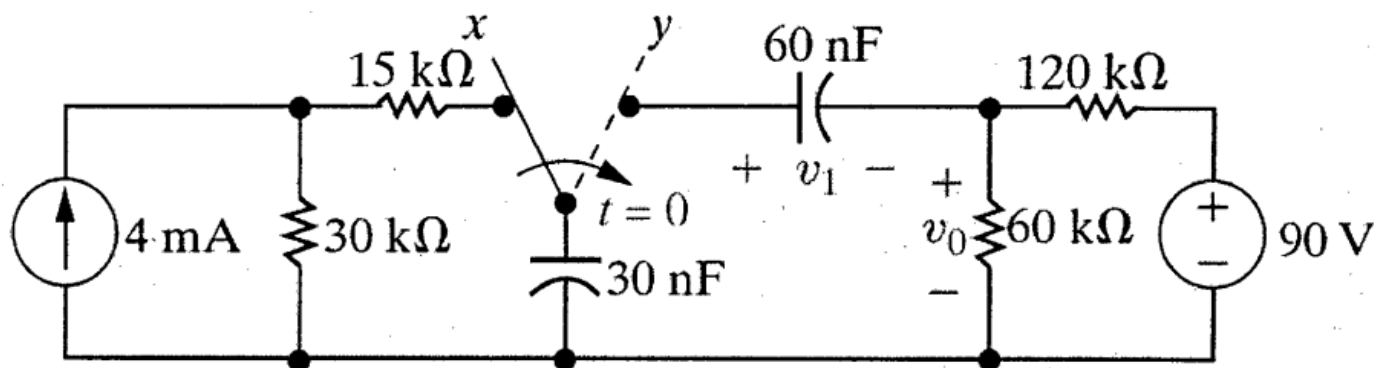


Inspiring Excellence

Department of CSE
CSE250: Circuits and Electronics
Assignment-2
Full Marks: 30

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1. The switch in the circuit in Figure has been in position x for a long time. The initial charge on the 60nF capacitor is zero. At $t=0$ the switch moves instantaneously to position y. Determine the value of $v_1(t)$ and the current of the $120\text{k}\Omega$ resistor for $t > 0$. (15)



2. The 240 V source in the circuit in Figure for Q. 2b is inadvertently short-circuited at its terminals a, b. At the time the fault occurs, the circuit has been in operation for a long time. How many microseconds after the short circuit has occurred is the current in the short equal to 114 A? (15)

